The even supersingular branches of Swisher's (E.3) and (F.3)

Abstract

We prove a one-step congruence modulo p^{3r} for two alternating truncated hypergeometric sums. Its two-step consequence gives the modulus p^{3r-2} of Swisher's even branches at the original cutoffs, including $p = 3$ for (F.3). The proof uses a terminating Dougall sum, quadratic interpolation and divisibility of every positive moment.

1 The alternating sums

Fix $d = 3$ or 4 and an odd prime $p \equiv -1 \pmod{d}$. Let a_r be the element of $\{1, d-1\}$ congruent to p^r modulo d. For $a \in \{1, d-1\}$ and $N = a + dn$ with $n \geq 0$, write

$$b_{a,k} = \frac{(a/d)_k}{k!}, \qquad S_a(N) = \sum_{k=0}^{n} (-1)^k (2dk + a) b_{a,k}^3, \qquad T_r = S_{a_r}(p^r).$$

Here $(x)_k$ is the rising factorial and $T_0 = 1$. The factor $2dk + a$ fixes the normalization for the companion family as well.

Theorem 1. *For $r \geq 1$ put $\epsilon_r = (-1)^{(pa_{r-1} - a_r)/d}$. Then*

$$T_r \equiv p\epsilon_r T_{r-1} \pmod{p^{3r}}.$$

Consequently, for every even $r \geq 2$,

$$\boxed{S_1(p^r) \equiv p^2 S_1(p^{r-2}) \pmod{p^{3r-2}}.}$$

For $d = 3$ this is the even supersingular (E.3) congruence, and for $d = 4$ it is the corresponding (F.3) congruence of Swisher's Table 2.

2 A terminating evaluation

Fix a, N and abbreviate $b_k = b_{a,k}$, $H_i = a + di$, $E_i = di$. Put

$$S_n(y) = \sum_{k=0}^{n} (-1)^k (2dk + a) b_k^3 \frac{\prod_{i=0}^{k-1} (1 - y/H_i^2)}{\prod_{i=1}^{k} (1 - y/E_i^2)}.$$

At $H = H_h$, $0 \leq h \leq n$, the terms beyond h vanish and

$$S_n(H^2) = (-1)^h H. \tag{1}$$

Indeed, put $\alpha = a/d$, $B = 2\alpha + h$. The sum divided by a is the terminating ${}_4F_3(-1)$ with upper parameters $\alpha, 1 + \alpha/2, B, -h$ and lower parameters $\alpha/2, 1 + \alpha - B, 1 + \alpha + h$. Dougall's ${}_5F_4(1)$ summation [4], with one upper parameter $-h$ and another tending to infinity, evaluates it as $(1 + \alpha)_h / (1 + \alpha - B)_h = (-1)^h (\alpha + h)/\alpha$. The limit is termwise in a finite sum.

3 Divisibility of all positive moments

Define $\mathcal{H}_a(N) = \{H_0, \dots, H_n\}$, $D_n(y) = \prod_{i=1}^n (1 - y/E_i^2)$ and $P_n(y) = \prod_{i=0}^n (1 - y/H_i^2)$. Since $D_n S_n$ is a polynomial of degree at most n, partial fractions and (1) give

$$F_n(y) = \frac{D_n(y)S_n(y)}{P_n(y)} = \sum_{H \in \mathcal{H}_a(N)} \frac{\mu_{a,N}(H)}{1 - y/H^2},$$

where

$$\mu_{a,N}(H) = (-1)^{(H-a)/d} H \frac{\prod_{i=1}^n (1 - H^2/E_i^2)}{\prod_{K \in \mathcal{H}_a(N),\, K \neq H} (1 - H^2/K^2)}.$$

The nodes are distinct and none is a positive multiple of d. Write $M_j(N) = \sum_H \mu_{a,N}(H) H^{2j}$, with a fixed. In particular $M_0(N) = F_n(0) = S_a(N)$.

Lemma 2. *For each $j \geq 0$ there is a polynomial $U_j \in \mathbb{Z}[1/2][X]$, depending on d, a, such that*

$$M_{j+1}(N) = (-1)^n (N b_n)^3 U_j(N), \qquad U_0 = 1.$$

If $N = p^r$, all the moments $M_j(N)$ with $j \geq 1$ belong to $p^{3r}\mathbb{Z}_p$.

Proof. For $n \geq 1$, separating the last term of S_n gives

$$(1 - y/N^2) F_n(y) = (1 - y/(N-a)^2) F_{n-1}(y) + (-1)^n (2N - a) b_n^3.$$

Expand $F_n(y) = -\sum_{j \geq 1} M_j(N) y^{-j}$ at infinity. The constant coefficient gives $M_1(N) = (-1)^n (N b_n)^3$ by induction: use $(N-d)b_{n-1} = (N-a)b_n$ and $M_1(a) = a^3$. The coefficient of y^{-j}, $j \geq 1$, gives

$$\frac{M_{j+1}(N)}{N^2} - M_j(N) = \frac{M_{j+1}(N-d)}{(N-a)^2} - M_j(N-d).$$

After normalization this recurrence requires

$$X U_j(X) + (X - a) U_j(X - d) = X^3 U_{j-1}(X) + (X - a)^3 U_{j-1}(X - d).$$

Over $R = \mathbb{Z}[1/2]$, take U_{j-1} invariant under $X \mapsto a - d - X$. The right side is odd about $a/2$. The operator on the left maps polynomials even about $(a-d)/2$ to polynomials odd about $a/2$. In the centered bases $(X - (a-d)/2)^{2i}$ and $(X - a/2)^{2i+1}$, its matrix is triangular with diagonal 2. It is invertible on each matched finite degree filtration over R. Starting with $U_0 = 1$ therefore gives every U_j in $R[X]$. At $X = a$ the recurrence gives $U_j(a) = a^{2j}$, as required by the single node for $n = 0$. Induction in j and then n gives the asserted formula. Since $-a/d \in \mathbb{Z}_p$ and $b_n = (-1)^n \binom{-a/d}{n} \in \mathbb{Z}_p$, it also gives the divisibility. Binomial polynomials take integral values on $\mathbb{Z}_p$ by continuity from the nonnegative integers. □

4 Transport through one prime-power level

Put $N = p^r$, $M = p^{r-1}$, $a = a_r$, $b = a_{r-1}$ and $e = (pb - a)/d$. Write $M = b + dm$; then $n = pm + e$ with $0 \le e < p$. The nodes divisible by p are exactly pJ for $J \in \mathcal{H}_b(M)$. The p-divisible denominator indices among $d, 2d, \dots, nd$ are $pd, 2pd, \dots, mpd$. All these statements include $m = 0$.

First remove the contribution of unit nodes. For $j_t = (p-1)p^t$, H^{2j_t} tends p-adically to 1 if $p \nmid H$ and to 0 otherwise. By Lemma 2 and the finiteness of the node set,

$$\sum_{p \nmid H} \mu_{a,N}(H) = \lim_{t \to \infty} M_{j_t}(N) \in p^{3r}\mathbb{Z}_p. \tag{2}$$

This limit is valid for rational weights of arbitrary individual valuation.

If $J = b + di$, then $(pJ - a)/d = pi + e$. Thus the ratio of the signed node evaluations (1) is $p(-1)^e$. Cancelling the factors with p-divisible indices in the weight formula yields

$$\mu_{a,N}(pJ) = p(-1)^e \mu_{b,M}(J) R(J), \qquad R(X) = \frac{\prod_{1 \le i \le n,\, p \nmid E_i} (1 - p^2X^2/E_i^2)}{\prod_{K \in \mathcal{H}_a(N),\, p \nmid K} (1 - p^2X^2/K^2)}.$$

All indices in R are units. Expanding its inverse factors geometrically,

$$R(X) = 1 + \sum_{j \ge 1} c_j X^{2j}, \qquad c_j \in p^{2j}\mathbb{Z}_p.$$

The series converges on $\mathbb{Z}_p$. Summing over the finitely many J and using Lemma 2 at the lower level shows that

$$p \sum_J \mu_{b,M}(J)(R(J) - 1) = p \sum_{j \ge 1} c_j M_j(M) \in p^{1+2+3(r-1)}\mathbb{Z}_p = p^{3r}\mathbb{Z}_p.$$

Here $M_j(M)$ uses the residue class b. Together with (2) this gives $T_r \equiv p(-1)^e T_{r-1} \pmod{p^{3r}}$.

At two consecutive levels the sign exponents add up to $((pb - a) + (pa - b))/d = p - 1$, since $a + b = d$. Their signs multiply to 1. The preceding step's error is multiplied by p, so the two-step precision is at least $\min(3r, 1 + 3(r-1)) = 3r - 2$. At every even level both endpoint sums have $a = 1$, as in Theorem 1. For $d = 4, p = 3, r = 1$, $a = 3$, $n = 0$, and $T_1 = 3 = pT_0$; all later levels are covered by the same argument.

References

[1] H. Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences 2 (2015), 18, Table 2.

[2] V. J. W. Guo and W. Zudilin, *Dwork-type supercongruences through a creative q-microscope*, J. Combin. Theory Ser. A 178 (2021), 105362, §3.4.

[3] V. J. W. Guo, *q-Analogues of two supercongruences in the (E.3) and (F.3) conjectures of Swisher*, Rocky Mountain J. Math. 55 (2025), 147–154. The precise overlap with its theorems has not been verified. This note makes no priority claim.

[4] NIST Digital Library of Mathematical Functions, Eq. 16.4.9, Rogers–Dougall very-well-poised sum.